Dot Product of Two Sparse Vectors (medium)

Problem Statement

Given two `sparse vectors`, efficiently compute the `dot product` of them.

A `sparse vector` is one in which most elements are zero.

Implement class `Solution`:

- `Solution(nums)` Initializes the object with the vector nums
- `dotProduct(vec)` Calculates the dot product between the `vec` and instance of `Solution`.

Examples

- **Example 1**:
  - **Input**: vec1 = `[1, 0, 0, 2]`, vec2 = `[2, 3, 0, 1]`
  - **Expected Output**: `4`
  - **Justification**: The dot product is `(1*2) + (0*3) + (0*0) + (2*1) = 2 + 0 + 0 + 2 = 4`.
- **Example 2**:
  - **Input**: vec1 = `[0, 0, 0, 0]`, vec2 = `[1, 1, 1, 1]`
  - **Expected Output**: `0`
  - **Justification**: Since `vec1` is all zeros, the dot product is `0` regardless of `vec2`'s values.
- **Example 3**:
  - **Input**: vec1 = `[1, 0, 0, 2, 3]`, vec2 = `[0, 3, 0, 4, 0]`
  - **Expected Output**: `8`
  - **Justification**: The dot product is `(1*0) + (0*3) + (0*0) + (2*4) + (3*0) = 8`.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    private int[] vector;

    // Constructor to store the entire vector
    public Solution(int[] nums) {
        // ToDo: Write Your Code Here.
    }

    // Method to calculate the dot product
    public int dotProduct(Solution vec) {
        int result = 0;
        // ToDo: Write Your Code Here.
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

```
[1,0,0,2,3]
[0,3,0,4,0]
```

Use Example Testcases | Run | Submit

← Previous

Solution: Nested List Weight Sum

Next →

Solution: Dot Product Of Two Sparse Vectors

Mark as Completed